

LIEGE AP, Belgium

WMO: 064780

Lat: 50.646N

Lon: 5.456E

Elev: 201

StdP: 98.93

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-4.8	-10.3	1.6	-4.6	-7.8	2.0	-3.4	13.4	8.6	12.3	8.2	2.9	50	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	29.9	20.2	27.6	19.2	25.6	18.5	21.1	27.7	20.1	25.9	19.1	24.4	3.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.0	24.4	18.0	13.2	22.8	17.1	12.5	21.7	62.0	27.8	58.4	26.0	55.1	24.4	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.5	8.4	DB	-9.3	33.8	3.2	1.8	-11.6	35.2	-13.4	36.2	-15.2	37.3	-17.5	38.6	(4)
(5)			WB	-9.7	23.3	3.1	1.4	-11.9	24.3	-13.7	25.2	-15.5	26.0	-17.7	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	3.0	3.7	6.6	10.0	13.6	16.6	18.8	18.4	15.1	11.4	6.9	3.8	(6)	
	DBStd	6.74	4.18	4.20	3.80	4.03	3.81	3.51	3.56	3.22	3.17	3.64	3.65	4.06	(7)	
	HDD10.0	899	218	177	118	49	9	0	0	0	1	27	104	195	(8)	
	HDD18.3	2922	476	408	365	252	153	76	38	38	107	216	343	451	(9)	
	CDD10.0	1152	1	2	11	48	122	198	271	261	154	71	12	2	(10)	
	CDD18.3	134	0	0	0	1	7	23	51	41	11	1	0	0	(11)	
	CDH23.3	1023	0	0	0	9	53	183	399	320	57	1	0	0	(12)	
	CDH26.7	290	0	0	0	1	10	47	122	101	9	0	0	0	(13)	
(14) Wind	WSAvg	4.0	4.8	4.6	4.4	3.8	3.6	3.5	3.5	3.4	3.6	4.1	4.3	4.8	(14)	
(15) Precipitation	PrecAvg	828	70	63	60	51	64	75	81	87	63	65	66	77	(15)	
	PrecMax	1068	143	143	136	127	158	149	266	256	185	176	109	173	(16)	
	PrecMin	615	6	7	5	1	20	12	35	19	15	32	4	21	(17)	
	PrecStd	131	32	31	29	28	32	36	47	55	41	29	25	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	14.8	19.0	24.8	28.0	31.3	33.1	32.6	28.0	22.8	17.2	13.2	(19)	
		MCWB	10.5	9.0	11.8	15.5	17.8	21.0	20.5	21.2	19.3	16.7	13.1	11.0	(20)	
	2%	DB	11.1	12.1	16.3	21.6	24.5	27.7	30.0	29.4	24.9	19.9	14.7	12.0	(21)	
		MCWB	9.5	8.6	10.8	13.9	16.6	19.4	20.3	20.1	18.2	15.4	12.1	10.2	(22)	
	5%	DB	9.7	10.5	14.0	18.9	22.4	25.2	27.4	27.0	22.4	18.0	13.3	10.7	(23)	
		MCWB	8.3	8.3	9.9	12.7	15.7	18.4	19.2	19.2	17.0	14.6	11.3	9.3	(24)	
	10%	DB	8.4	9.3	12.1	16.4	20.2	23.1	25.3	24.4	20.2	16.4	11.9	9.3	(25)	
		MCWB	7.0	7.5	9.0	11.5	14.7	17.5	18.5	18.1	15.9	13.8	10.3	8.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.8	13.1	16.9	19.7	22.1	22.5	22.9	20.3	17.2	14.0	11.7	(27)	
		MCDB	12.1	13.0	17.5	23.6	25.8	28.3	30.0	30.5	25.8	20.9	16.1	12.8	(28)	
	2%	WB	9.6	9.5	11.5	14.6	17.9	20.4	21.0	21.0	18.7	16.0	12.5	10.5	(29)	
		MCDB	10.7	11.2	14.7	19.7	23.0	26.1	28.0	27.4	23.7	19.0	14.3	11.7	(30)	
	5%	WB	8.4	8.6	10.4	13.1	16.4	19.0	20.0	19.7	17.6	15.0	11.5	9.4	(31)	
		MCDB	9.5	10.2	13.2	17.8	20.8	24.0	25.8	25.3	21.5	17.3	13.1	10.6	(32)	
	10%	WB	7.1	7.5	9.4	11.9	15.2	17.9	19.0	18.6	16.5	14.1	10.4	8.1	(33)	
		MCDB	8.3	9.2	11.8	16.0	19.3	22.3	24.5	23.2	19.7	16.2	11.8	9.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	5.1	6.7	8.1	8.4	8.7	8.6	8.4	7.5	6.2	4.6	4.0	(35)	
		MCDBR	4.9	6.4	9.5	11.1	11.3	12.0	11.8	12.3	10.1	7.7	5.7	4.5	(36)	
	5% WB	MCWBR	4.4	4.6	5.7	6.1	5.9	6.1	5.1	5.5	5.0	4.6	4.2	3.8	(37)	
		MCWBR	4.9	5.6	7.9	10.0	9.9	10.5	10.7	10.7	9.1	7.0	5.2	4.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.313	0.336	0.368	0.401	0.410	0.410	0.412	0.405	0.376	0.353	0.328	0.311	(40)	
		taud	2.427	2.383	2.306	2.238	2.252	2.269	2.261	2.307	2.373	2.433	2.430	2.398	(41)	
	Ebn at Noon		719	793	827	838	847	848	839	826	812	763	690	657	(42)	
		Edh at Noon	61	82	105	126	130	128	128	116	98	77	61	55	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.78	1.45	2.60	4.13	4.91	5.35	5.08	4.32	3.26	1.93	0.92	0.61	(44)	
	RadStd		0.11	0.29	0.37	0.48	0.48	0.48	0.61	0.46	0.34	0.16	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.49	N/A	N/A	N/A	N/A	(46)	
(47) Regional (80 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page